

PayWise Owner's Personal Guide & Viva Preparation

This guide is written in plain English for explaining the project confidently to teachers and evaluators, even without prior Flutter experience.

1. Flutter in 10 Minutes (Plain English)

- **Widget:** A Lego brick. Every single visual element (text, button, image, container) is a widget.
 - **Widget Tree:** How widgets are stacked together. A screen is just a tree of Lego bricks inside a parent brick.
 - **State:** The information the app remembers right now (like your wallet balance or if you are logged in).
 - **build():** The factory method where Flutter draws the Lego blocks based on the current state.
 - **Scaffold:** The standard blank mobile page template with a top bar, body area, and bottom navigation bar.
 - **Column / Row:** Layout bricks that line up widgets vertically (Column) or horizontally (Row).
 - **ListView:** A scrollable vertical column that only builds items as you scroll to them.
 - **Provider:** A megaphone. When data changes in Firestore, Provider announces it so all screens refresh instantly.
 - **async / await:** Asking for something that takes time (like talking to Firebase) without freezing the app while waiting.
 - **Firestore Stream:** A live pipeline. When anything changes in the database, the new data flows straight into the app.
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2. How the App Starts (lib/main.dart)

1. **main()** runs first. It initializes Flutter system hooks and connects Firebase using **DefaultFirebaseOptions**.
 2. **ChangeNotifierProvider** wraps the root with **AppProvider**.
 3. **PayWiseApp** sets the light theme and hands navigation over to **appRouter** (configured with **GoRouter**).
 4. If a user is not signed in, **GoRouter** redirects to **/login (LoginScreen)**.
 5. Once signed in (via email or "Continue as demo user"), the router opens the main 5-tab shell (/).
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3. Where Things Live (Pointer Table)

Feature	File to Open	What It Does
Pricing & Limit Rules	lib/core/pricing_rules.dart	Daily ■1,00,000 bank transfer limit, 2 free credit card payments then 1% fee
Category Colors & Icons	lib/core/categories.dart	8 spending categories with distinct color tokens and line icons
Theme & Tokens	lib/core/theme/	Exact colors, font scale (Plus Jakarta Sans), and light theme
Database Gateway	lib/data/firestore_service.dart	The only file that communicates with Cloud Firestore

Demo Seed Data	<code>lib/data/demo_data.dart</code>	Generates 40 transactions, 8 contacts, 5 bills, 2 cards, and holdings
Auth & Anonymous Login	<code>lib/data/auth_service.dart</code>	Handles email sign-in and instant demo login
Login Screen	<code>lib/features/auth/login_screen.dart</code>	Stitch-matched auth screen with demo access button
5-Tab Shell & Scan Button	<code>lib/features/home/nav_shell.dart</code>	Bottom bar with Home, History, Wealth, Services, Rewards + center Scan button
History & Donut Chart	<code>lib/features/history/history_screen.dart</code>	Interactive <code>fl_chart</code> donut and transaction audit sheet
Bills & Autopay	<code>lib/features/bills/bills_screen.dart</code>	Utility bill payment with Autopay frequency selector
Credit Card 1% Fee Rule	<code>lib/features/bills/credit_card_bill_screen.dart</code>	Card settlements with automatic fee preview
Wealth & Digital Gold	<code>lib/features/wealth/digital_gold_screen.dart</code>	Live ticking 24K gold rate with instant buy/sell
Insurance Discovery	<code>lib/features/insurance/insurance_screen.dart</code>	Policy selector and 4-step claim status progress
Lending & Live EMI	<code>lib/features/lending/lending_screen.dart</code>	Interactive loan slider with live EMI calculation
Rewards & Scratch Cards	<code>lib/features/wallet/rewards_screen.dart</code>	Scratch card reveal animations and voucher redemption
Profile & Debug Panel	<code>lib/features/profile/profile_screen.dart</code>	QR modal, failure toggle, and session logout

4. Viva Cheat Sheet (Top Questions & Honest Answers)

1. Why Flutter instead of React Native or Native Android?

Flutter compiles directly to native ARM machine code and WebAssembly/CanvasKit with zero runtime JavaScript bridge, ensuring 60fps animations and pixel-identical rendering on both Android and Web.

2. Why Firebase Firestore over traditional SQL?

Firestore is a document database offering real-time streaming listeners (`snapshots()`). When a payment completes or gold is purchased, the UI updates instantly across all open screens without manual polling or page reloads.

3. Why Provider for state management instead of Bloc or Riverpod?

Provider is Flutter's official, recommended state management solution. It relies on standard Dart `ChangeNotifier`, making it easy to reason about, maintain, and test without unnecessary boilerplate.

4. Is real money involved?

No. All financial transactions, UPI simulations, and gold purchases are realistically simulated and persisted in Google Cloud Firestore.

5. How are pricing rules enforced?

All rules are centralized in `lib/core/pricing_rules.dart` and backed by automated unit tests in `test/pricing_test.dart`.

6. How does the web app stay phone-sized on a desktop monitor?

`NavShell` wraps all pages inside a `ConstrainedBox(maxWidth: 430)` centered on screen, retaining a realistic smartphone aspect ratio.

7. How is code readability and maintainability ensured?

Every single **.dart** file is kept under 150 lines, and complex widgets are decomposed into reusable sub-widgets.

5. Live Demonstration Script for Examiners (5 Minutes)

1. Step 1 - Authentication:

- Open <https://paywise-ae977.web.app>.
- Click "Continue as demo user" to sign in instantly with a seeded demo account.

2. Step 2 - Quick Payment & PayPause Friction:

- Tap "To Mobile / Contact" on the Home dashboard.
- Choose a contact (e.g., "Aarav Sharma").
- Type **2500** into the custom numpad and tap Pay.
- Show the examiner the **PayPause Mindful Sheet**: explain that any payment $\geq$ \$2,000 triggers a 3-second reflection pause showing month-to-date category spend.
- Confirm payment to see the green checkmark success screen.

3. Step 3 - Real-Time History & Donut Chart:

- Tap the "History" tab in the bottom bar.
- Point out the interactive Donut chart (**fl_chart**) and the newly inserted transaction at the top of the list.
- Tap the transaction to reveal the detailed UTR audit bottom sheet.

4. Step 4 - Credit Card Fee Rule:

- Tap "Services" to "Credit Card Bill".
- Show the fee badge: first 2 card payments are free, subsequent card payments calculate a 1% convenience fee automatically.

5. Step 5 - Digital Gold Live Ticker:

- Tap "Wealth" to "Digital Gold".
- Show the ticking live gold rate ($\text{₹}7,420/\text{g}$ with market fluctuation) and buy $\text{₹}500$ of gold.
- Observe the vault balance increment immediately.

6. Step 6 - Profile & Debug Tools:

- Tap the top-left profile avatar to open </profile>.
- Show the personal UPI QR preview modal and explain the failure simulation toggle for testing resilience.